

Dhruva Ramachandran

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science | GPA: 3.1/4

August 2025 – May 2028

Relevant Courses: CS 25100: Data Structures & Algorithms, MA26500: Linear Algebra, CS24000: C Programming, CS18200: Foundations of Computer Science (Discrete Math), CS 25000: Computer Architecture, MA26600: Differential Equations

Monta Vista High School

Cupertino, CA

Aug. 2021 – June 2025

EXPERIENCE

Software Developer

September 2025-May 2026

Hack the Future

West Lafayette, IN

- **Built a full-stack volunteer tracking platform** for an agricultural non-profit organization, using MongoDB, Next.js, and React supporting **300+** users
- Architected and implemented real-time event sign ups and email notifications via Nodemailer, **increasing communication efficiency**
- Worked with teammates to build inventory tracking service, reducing manual workload for admins

PROJECTS

Agentic Web Automation Tool | *Playwright, FastAPI, LLM* | Github Repo

June 2026

- Architected an autonomous **multi-service agentic pipeline (FastAPI microservices)** that automates end-to-end internship discovery, evaluation, and application submission
- Engineered a **Playwright scraping service** with persistent session auth to extract Handshake listings, integrated with an **LLM analysis engine (Groq/Llama 3.3 70B)** to parse key skills and employer values
- Designed a **database-backed architecture (SQLite, REST APIs)** to dynamically personalize resumes and automate submission flows tailored to extracted listing criteria

PubMed RAG Chatbot | *Retrieval Augmented Generation, LLM* | Github Repo

March 2026 – May 2026

- Engineered a **full-stack agentic RAG application** to query PubMed literature, connecting a React frontend, Express/TypeScript backend, and Python FastAPI retrieval service
- Architected a **hybrid retrieval-augmented generation pipeline** combining real-time PubMed API retrieval, vector semantic search, and LLMs for natural language queries
- Containerized and deployed the multi-service architecture using **Docker Compose** across Railway and Vercel with environment-based configuration management

Invert | *iOS, Swift, SwiftUI* | Github Repo

January 2024 - April 2024

- Built a native **iOS productivity app** in Swift/SwiftUI with a modular MVVM architecture separating data models, business logic, and UI components
- Implemented a multi-state timer system (start/resume/stop) with real-time audio feedback and **device motion sensor integration** to detect device pick-ups
- Applied and got app added to the App Store, and promoted via Instagram and scaled to **60+ downloads** in the first week of release

TECHNICAL SKILLS

Languages: Java, Python, Swift, C, Python

Tools & Platforms: Git, Claude Code, Vercel, Docker, XCode

Frameworks: FastAPI, React, Node.js, Flask, Next.js, MongoDB, Playwright

CERTIFICATIONS

Retrieval Augmented Generation (RAG): DeepLearning.ai, March 2026

Certificate of Achievement: Research, Design and Development for Global Good: Foothill College, June 2024